EARL LOUIS

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EDUCATION

University of Connecticut — **Expected December 2026**
Bachelor of Science in Robotics Engineering (Electronics Track)

EXPERIENCE

Barnes Aerospace — **East Granby, CT**
Control Systems Intern, Smart Connected Factory — June 2025 - Present
- Designed the instrumentation and control scheme for an upgraded industrial fluid handling process, authoring the P&ID and control panel covering level and temperature sensing, immersion heating, and pump interlocks
- Designed a plant safety sensor network using wet switches and normally closed contacts to detect spills, eyewash and AED use, and enclosure access, with alarms automatically dispatching to the appropriate emergency response teams
- Integrated a pH probe and flowmeters into the plant control network for continuous environmental monitoring, running field testing, troubleshooting, and startup to bring discharge monitoring into continuous compliance
- Built operator-facing SCADA displays in Ignition Perspective showing live machine state, process values, and OEE and KPI trending, giving the plant real-time visibility into how every connected machine is running
- Connected CNC machines and industrial equipment to the plant IoT network over Modbus TCP, EtherNet/IP, MTConnect, and OPC UA, reviewing PLC I/O and machine documentation to determine how each asset exposes data

Norwich Public Utilities — **Norwich, CT**
Electrical Engineering Intern — June 2024 - January 2025
- Designed 2 alternative distribution layouts to resolve a residential voltage drop blocking EV charging, specifying transformer rating, customer split, and secondary run length for each, and delivered both for senior engineer review
- Tested 41 newly acquired pole-top transformers by turns ratio, documenting results against tolerance specifications and flagging out-of-tolerance units for follow-up
- Investigated distribution faults by pulling relay event records to identify the faulted phase and walking the affected circuit, surfacing a discrepancy between the system one-line and the as-built field configuration

PROJECTS

UAV Health Monitoring System | Python, Raspberry Pi, FFT, SPI, I2C, UART, Betaflight, RF Telemetry
- Built a Raspberry Pi based diagnostics add-on for a quadcopter, running 6 concurrent Python threads to read 4 accelerometers, 4 current sensors, and 2 temperature sensors while pulling live motor RPM from a Betaflight flight controller
- Compared vibration and current draw per motor against a baseline that adjusts with motor speed, and flagged battery heating faster than the current draw predicted, catching real faults without false alarms as throttle changes
- Localized an injected fault to the correct motor during bench testing, dropping its health score from above 90% to roughly 25% within seconds with no false positives on the 3 healthy motors

Wavefront Path Planning Robot | Python, OpenCV, Arduino, Romi 32U4
- Wrote the perception and planning software for a wheeled robot navigating to a clicked goal using only an overhead camera, tracking position and heading from 2 color markers and replanning each second so obstacles can be moved mid-run
- Planned routes around obstacles on a safety-padded occupancy grid map of the room and smoothed them into a curve the robot could actually drive, adding extra padding at corners after testing showed it clipping them

Sound Intensity Visualizer | AVR, Bare-Metal C, I2C, SPI
- Built a microcontroller system that samples sound from a microphone and shows live intensity on an LED matrix and an OLED readout, with modes for current level, scrolling waveform, and recorded playback
- Wrote the OLED display driver directly from its datasheet with no external library, buffering the screen in memory so text updates did not tie up the data bus

Can-Opening Robot Arm | Autodesk Fusion, 3D Printing, Arduino, I2C
- Built a 4-DOF robotic arm that opens a soda can automatically when one is placed in its holder, aimed at users with limited grip strength
- Designed and 3D printed the full structure in CAD across 2 to 3 revisions per part, including a hooked end effector shaped to slip under a tab, and programmed a 5-phase opening motion with each servo calibrated to its joint travel

SKILLS

Languages and Libraries: Python, C, C++, MATLAB, NumPy, OpenCV
Tools: ROS, Simulink, Autodesk Fusion, KiCad, LTspice, AutoCAD, Ignition (8.1 and 8.3 credentials), Git, Linux
Protocols and Hardware: I2C, SPI, UART, Modbus TCP, EtherNet/IP, OPC UA, MTConnect, Raspberry Pi, Arduino, AVR, Oscilloscope, Soldering